ColdChain AI — All Models Explained

Every model from the syllabus, in simple words: what it is, what score it gave, and why it was chosen or rejected.

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Note (Oct 4, 2026): Checked line-by-line against the official Data Science-I syllabus (25MCA3001, from the DDU GNUMS portal). Every model below appears in the syllabus — nothing outside it is used.

Syllabus models used: Linear Regression, Logistic Regression, KNN, Naive Bayes, Decision Tree, Random Forest, SVM (both regression and classification versions where they apply). Winners: **Linear Regression** (regression) and **Decision Tree** (classification).

Baseline-persistence Task: regression

✔ USED AS FINAL MODEL

What is this model? (simple words)

A dumb rule — predict that next hour temperature will be the same as this hour's temperature.

What output did I get when I used it?

Tried: none — this is a baseline, not a trained model.
Best setting: -
Result: RMSE 1.154, MAE 0.8104, R2 0.3997.

Why did I not use it? / What is the issue if I use it?

Good points: Needs no training. Very easy to understand.
Problem: It cannot predict any change. It just repeats the current value.
baseline only — it exists so we can check that our real models actually learned something.

LinearRegression Task: regression

✔ USED AS FINAL MODEL

What is this model? (simple words)

It draws a straight line through the data. The simplest model in our syllabus.

What output did I get when I used it?

Tried: none — fit directly. LinearRegression has no hyperparameters, so running a grid search on it would be a fake search.
Best setting: {} (no tuning)
Result: RMSE 1.0374, MAE 0.7046, R2 0.5149. Training time: 0.0 s.

Why did I not use it? / What is the issue if I use it?

Good points: Very easy to explain in viva. Trains instantly.
Problem: Can only learn straight-line relations. If two features say the same thing, the weights become unstable.
CHOSEN — lowest RMSE of every syllabus model (1.0374), trains instantly, and it is the first topic in Unit 2 of the official syllabus.

KNNRegressor Task: regression

✘ NOT CHOSEN

What is this model? (simple words)
To predict, it finds the k most similar hours from training data and averages their temperatures.

What output did I get when I used it?
Tried: n_neighbors = [3, 5, 9, 15, 25, 35] with GridSearchCV.
Best setting: n_neighbors = 25
Result: RMSE 1.0941, MAE 0.7322, R2 0.4604. Training time: 0.5 s.

Why did I not use it? / What is the issue if I use it?
Good points: No assumptions about the data shape. Simple idea, easy to explain.
Problem: Slow at prediction time on big data (it compares with every training row). RMSE 1.0941 is clearly worse than LinearRegression.
NOT CHOSEN — worse score than the linear models, and it does not scale well.

DecisionTreeReg Task: regression

✘ NOT CHOSEN

What is this model? (simple words)
It splits the data with yes/no questions (like "is door_open_min > 10?") until it can predict the temperature.

What output did I get when I used it?
Tried: max_depth = [3, 5, 8, 12, None], min_samples_split = [2, 5, 10] with GridSearchCV.
Best setting: max_depth = 3, min_samples_split = 2
Result: RMSE 1.107, MAE 0.7681, R2 0.4476. Training time: 0.8 s.

Why did I not use it? / What is the issue if I use it?
Good points: Very easy to draw and explain. Needs no feature scaling.
Problem: Worst score of all trained models (RMSE 1.107). Single trees overfit easily.
NOT CHOSEN — the worst regressor we tried.

RandomForestReg Task: regression

✘ NOT CHOSEN

What is this model? (simple words)
Many decision trees trained on random parts of the data, then their predictions are averaged.

What output did I get when I used it?
Tried: n_estimators = [50, 100, 200], max_depth = [5, 10, 15, None], min_samples_split = [2, 5] with RandomizedSearchCV (10 tries).
Best setting: n_estimators = 200, min_samples_split = 5, max_depth = 5
Result: RMSE 1.0612, MAE 0.7228, R2 0.4923. Training time: 29.0 s.

Why did I not use it? / What is the issue if I use it?
Good points: Handles non-linear relations. More stable than a single tree.
Problem: Slowest model we trained (29 seconds for 200 trees). Still worse than LinearRegression. Hard to explain tree-by-tree in viva.
NOT CHOSEN — slower and worse than LinearRegression.

SVR Task: regression

✘ NOT CHOSEN

What is this model? (simple words)
Support Vector Regression — it tries to fit a tube around the data and ignores small errors inside the tube.

What output did I get when I used it?
Tried: C = [0.1, 1, 10], gamma = ["scale", 0.01, 0.1] with GridSearchCV.
Best setting: C = 10, gamma = 0.01
Result: RMSE 1.0573, MAE 0.642, R2 0.496. Training time: 3.8 s.

Why did I not use it? / What is the issue if I use it?
Good points: Good at non-linear patterns. Interesting — it got the lowest MAE of all models (0.642).
Problem: RMSE (our selection metric) is worse than LinearRegression's. Sensitive to C and gamma, and needs scaled features.
NOT CHOSEN — lowest MAE is nice, but our rule picks on RMSE, and LinearRegression wins there.

Baseline-majority Task: classification

✔ USED AS FINAL MODEL

What is this model? (simple words)
A dumb rule — always predict "no excursion" (the majority class).

What output did I get when I used it?
Tried: none — this is a baseline, not a trained model.
Best setting: -
Result: accuracy 0.8715, precision 0.0, recall 0.0, F1 0.0.

Why did I not use it? / What is the issue if I use it?
Good points: Needs no training.
Problem: It catches zero excursions, so it is useless for our actual goal.
baseline only — it proves we must judge models by F1, not accuracy.

LogisticRegression Task: classification

✘ NOT CHOSEN

What is this model? (simple words)
It draws a straight boundary between "excursion" and "no excursion" and gives a probability.

What output did I get when I used it?
Tried: C = [0.1, 1.0, 10.0] with GridSearchCV, scoring on F1. Trained with class_weight="balanced".
Best setting: C = 0.1
Result: accuracy 0.8264, precision 0.3839, recall 0.5811, F1 0.4624. Training time: 0.3 s.

Why did I not use it? / What is the issue if I use it?
Good points: Fast and easy to explain. Gives probabilities, not just yes/no.
Problem: Can only learn a straight boundary. F1 0.4624 is below our winner.
NOT CHOSEN — decent, but DecisionTreeClf beat it on F1.

KNNClassifier Task: classification

✘ NOT CHOSEN

What is this model? (simple words)

To classify, it looks at the k most similar hours and takes a majority vote.

What output did I get when I used it?

Tried: k = [3, 5, 9, 15, 25, 35] with a manual CV loop — inside each fold the minority class was oversampled, so the validation scores stay honest.
Best setting: n_neighbors = 35
Result: accuracy 0.7882, precision 0.3261, recall 0.6081, F1 0.4245. Training time: 1.2 s.

Why did I not use it? / What is the issue if I use it?

Good points: Highest recall of all models (0.6081) — it catches the most excursions. Simple idea.
Problem: Lowest accuracy (0.7882) and lowest precision — it raises too many false alarms. F1 0.4245 is well below the winner.
NOT CHOSEN — catches a lot but cries wolf too often.

NaiveBayes Task: classification

✘ NOT CHOSEN

What is this model? (simple words)

It uses Bayes theorem and assumes every feature is independent of the others.

What output did I get when I used it?

Tried: none — nothing to tune, so we fit it directly on the oversampled training data (a grid search here would be a fake search).
Best setting: {} (no tuning)
Result: accuracy 0.7986, precision 0.325, recall 0.527, F1 0.4021. Training time: 0.0 s.

Why did I not use it? / What is the issue if I use it?

Good points: Trains instantly. Works okay on small data.
Problem: The independence assumption is wrong here (temp and humidity move together). Lowest F1 of all trained models.
NOT CHOSEN — weakest classifier we tried.

DecisionTreeClf Task: classification

✔ USED AS FINAL MODEL

What is this model? (simple words)

It splits the data with yes/no questions (like "is temp > 7.5?") until it can say excursion or not.

What output did I get when I used it?

Tried: max_depth = [3, 5, 8, 12, None], min_samples_split = [2, 5, 10] with GridSearchCV, scoring on F1. Trained with class_weight="balanced".
Best setting: max_depth = 3, min_samples_split = 2
Result: accuracy 0.8889, precision 0.5862, recall 0.4595, F1 0.5152. Training time: 0.9 s.

Why did I not use it? / What is the issue if I use it?

Good points: Highest F1 of all models (0.5152, tied with RandomForest). Very easy to draw and explain in viva. Fast.
Problem: Single trees can overfit — but here the best tree was shallow (depth 3), so it stayed simple. Recall 0.4595 means it still misses about half the excursions.
CHOSEN — tied RandomForestClf on F1, and our tie-break rule says the simpler model wins.

RandomForestClf Task: classification

✘ NOT CHOSEN

What is this model? (simple words)
Many decision trees trained on random parts of the data, then they vote.

What output did I get when I used it?
Tried: n_estimators = [50, 100, 200], max_depth = [5, 10, 15, None], min_samples_split = [2, 5] with RandomizedSearchCV (10 tries), scoring on F1. Trained with class_weight="balanced".
Best setting: n_estimators = 100, min_samples_split = 2, max_depth = 15
Result: accuracy 0.8889, precision 0.5862, recall 0.4595, F1 0.5152. Training time: 12.0 s.

Why did I not use it? / What is the issue if I use it?
Good points: Same best F1 as the decision tree (0.5152). Usually more stable than a single tree.
Problem: 100 trees take 12 seconds to train vs 0.9 for one tree. Much harder to explain in viva.
NOT CHOSEN — tied the decision tree on F1, but the tie-break rule picks the simpler model.

SVC Task: classification

✘ NOT CHOSEN

What is this model? (simple words)
Support Vector Classifier — it finds the boundary with the biggest gap between the two classes.

What output did I get when I used it?
Tried: C = [0.1, 1, 10], gamma = ["scale", 0.01, 0.1] with GridSearchCV, scoring on F1. Trained with class_weight="balanced".
Best setting: C = 10, gamma = 0.01
Result: accuracy 0.8611, precision 0.4651, recall 0.5405, F1 0.5. Training time: 2.6 s.

Why did I not use it? / What is the issue if I use it?
Good points: Close to the winner (F1 0.5). Good balance of precision and recall.
Problem: Still below 0.5152. Sensitive to C and gamma, needs scaled features, harder to explain than a tree.
NOT CHOSEN — second best, but the rule picks the highest F1.

Final choice

Regression (predict next-hour temperature): Linear Regression — RMSE 1.0374, R^2 0.5149. Beat the persistence baseline (1.154) and every other syllabus model; trains instantly; first topic of Unit 2 in the official syllabus.

Classification (predict excursion): Decision Tree — accuracy 0.889, F1 0.515. Tied RandomForest on F1; the simpler model wins the tie (1 tree vs 100 trees, faster, easier to explain).

Honest limits: the classifier still misses about half the real excursions (recall 0.46); regression R^2 0.51 is student-grade, not production-ready.